

Determining Minimum Separation Boundary for UAM Tactical Deconfliction

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BIOGRAPHY

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ABSTRACT

Urban air mobility (UAM) is an emerging paradigm in future mobility that utilizes the airspace above urban areas and incorporates innovative technologies to provide safe and efficient transportation. However, as UAM reaches maturity, the increased density in constrained airspace will raise safety concerns, particularly the risk of mid-air collisions (MAC). This emphasizes the importance of deconfliction in ensuring safe separations from other aircraft and static obstacles. Deconfliction includes strategic and tactical parts, which refer to pre-flight and in-flight deconfliction, respectively. Conventional aircraft perform deconfliction using sufficiently spaced out separation in wide airspace. Applying these extensive separation boundaries to UAM, however, could significantly compromise airspace efficiency. Therefore, it is necessary to determine a tightened separation boundary that enhances airspace efficiency while still meeting the safety requirement. Building on our previous research, which introduced a concept for a minimum separation boundary focusing on the strategic deconfliction, this study extends the scope to include the tactical deconfliction. The tactical deconfliction utilizes the detect-and-avoid (DAA) system to mitigate collisions when the strategic deconfliction fails. This study proposes a methodology for determining the minimum separation boundary for the tactical deconfliction, focusing on two key parameters representing the performance of the DAA system. Additionally, a case study was conducted by applying the proposed methodology to an airborne collision avoidance system X (ACAS X), a promising DAA system suitable for UAM. The results demonstrate that implementing the tactical deconfliction could reduce the separation boundary by 12% compared to scenarios without it.

Keywords: urban air mobility, minimum separation, safety risk assessment, tactical deconfliction, well-clear

1.0 INTRODUCTION

In urban and suburban areas, rapid population growth combined with limited infrastructure is giving rise to transportation challenges. Urban air mobility (UAM), a next-generation transportation, leverages airspace and incorporates innovative technologies to provide safe and efficient transportation (FAA, 2023). According to the FAA's UAM ConOps, the UAM system coexists with the national air space (NAS) and is designed to operate on designated routes. As the UAM system reaches maturity, traditional manned air traffic control (ATC) will become overloaded with the increased traffic volume, necessitating the development of an autonomous safety monitoring system to support multiple UAM operations. Hence, autonomous flight path management (FPM), responsible for ensuring vehicle separation and route adjustments, is essential for the safety of UAM operations (NASA, 2021).

FPM performs deconfliction that predicts and resolves the loss of separation with other traffic or static obstacles, such as mountains and buildings. In essence, it consists of strategic and tactical parts which refer to pre-flight and in-flight deconfliction respectively, and establishes a separation boundary around the trajectory ensuring there are no overlaps between the boundary and hazards. However, due to system uncertainties, these separation boundaries may fail to bound the actual position of an aircraft, resulting in what is known as a separation violation (SV) (Kim et al., 2014). An SV carries the potential for mid-air collisions (MAC), which can lead to severe casualties. Therefore, it becomes imperative to determine separation boundaries that guarantee the probability of SV to lower than the SV risk requirement. This SV risk requirement is derived from a target level of safety (TLS). TLS is a safety requirement for UAM operations defined as fatal accidents per hour. To ensure the system safety, TLS is allocated to each subsystem that can potentially cause SV, which is then converted to the SV risk requirement. Each subsystem should satisfy the SV risk requirement by reducing the subsystem's uncertainty or establishing the separation

boundary to mitigate the uncertainty. The separation boundary denotes a confidence interval for total system errors (TSE) (Kim et al., 2023).

Previous works (FAA, 2019; Vila et al., 2023; Reich, 1997; Chen et al., 2019) related to separation boundaries have primarily focused on assessing the safety risks associated with given separation boundaries. These boundaries are designed for general aviation (GA) or unmanned aircraft systems (UAS) and are typically large, making them impractical for UAM airspace. For instance, GA guidelines require maintaining a horizontal separation of 5 nautical miles and a vertical separation of 1,000 to 2,000 feet (FAA, 2019). UAS guidelines recommend a horizontal separation of 4,000 feet and a vertical separation of 450 feet (Chen et al., 2019). Although these extensive separation boundaries can readily meet the safety requirements, they can reduce airspace efficiency, particularly in constrained airspaces at low altitudes. To explore an optimal separation boundary, it is necessary to determine the smallest allowable separation boundary that ensures to meet the safety standards, which we refer to this as the “minimum separation boundary.”

In our previous research (Kim et al., 2023), we introduced a methodology for determining the safety-assured minimum separation boundary for UAM operations. This methodology involves identifying the TLS allocation scheme, conducting safety risk assessments (SRA), and analyzing system integrity. Firstly, TLS, a safety requirement, is distributed across subsystems of UAM that can cause SV, and each subsystem must guarantee its allocated TLS. For the SRA, the prior research established a safety risk model that quantifies the probability of SV resulting in a series of collision and fatal accidents assuming the worst-case scenario for the strategic deconfliction. The probability of SV is calculated based on the TSE distribution. The distribution is derived from the integrity risk analysis on the subsystem and its operational conditions, such as nominal and sensor failures. This methodology offers a systemic approach that reduces the need for extensive flight data without compromising conservatism. The results showed the possibility of significant reductions in separation boundaries while still satisfying the safety requirements compared to existing standards. Our previous research primarily focused on UAM separation scenarios involving only strategic deconfliction, because tactical deconfliction for UAM is still an area of ongoing research, and no standardized concept has been established yet. However, this may lead to an unnecessary overestimation of risk when the tactical deconfliction is implemented.

Tactical deconfliction is essential for UAM operations as a safety strategy to reduce collision risks when the strategic deconfliction fails. It utilizes a detect-and-avoid (DAA) system, which detects potential intruders and executes avoidance maneuvers sequentially (Johnson & Larrow, 2020). In an extension of our prior research, this paper proposes a methodology for determining the safety-assured minimum separation boundary for both strategic and tactical deconfliction. We have developed an improved safety risk model to consider the tactical deconfliction. This model describes the collision scenario handled by the DAA procedure, involving the collision risk in case of detection failure and avoidance failure despite correct detection. To parameterize the performance of DAA in a general way, we define two parameters: detection probability and avoidance risk ratio. The detection probability varies depending on the surveillance system and detection algorithm, while ARR varies depending on factors such as the avoidance algorithm and the maneuverability of the aircraft. As a result, we can derive the minimum separation boundary for the tactical deconfliction quantifying the safety enhancement offered by the DAA system.

A case study was conducted by applying our methodology to the airborne collision avoidance system X (ACAS X), a promising DAA system suitable for UAM (EUROCONTROL, 2022). Based on the simplified ACAS X configuration, we defined a “surveillance uncertainty” that represents the detection performance, and utilized it to calculate the probability of detection. In addition, a sensitivity analysis was performed to analyze the impact of surveillance uncertainty and ARR on the minimum separation boundary. This analysis then presented a performance criterion for reducing the separation boundary.

The structure of this paper is as follows: Section 2 introduces the strategic and tactical deconfliction and defines the separation boundary; Section 3 provides an overview of our previous research on the safety-assured minimum separation boundary; Section 4 describes our methodology for determining the minimum separation boundary for the tactical deconfliction; Section 5 presents a case study in which the proposed methodology is applied to the ACAS X, and conclusions are given in Section 6.

2.0 Strategic and Tactical Deconfliction in Urban Air Mobility

The single european sky air traffic management (ATM) research (SESAR) and NASA have proposed a Barrier Model for preventing airborne collisions in UAS, which is also suitably adapted for UAM (Vila et al., 2023; Johnson & Larrow, 2020). This model comprises three layers of deconfliction: Strategic, tactical, and collision avoidance (CA) (Manfredi & Jestin, 2018). Each layer includes an activation threshold (barrier) and evasive maneuvers. Note that while there are other definitions and variations within the barrier model, the one presented here is a basic and common interpretation.

- **Strategic Deconfliction:** In this layer, the well clear threshold (WCT) serves as the separation boundary. Strategic deconfliction involves long-term monitoring of the flight path, either before or during the flight. If a potential intruder is detected, adjustments to the flight path or arrival time are implemented.
- **Tactical Deconfliction:** This layer is activated when the strategic separation boundary, WCT, is violated by intruders. This layer monitors the flight path at short intervals during flight (less than 5 minutes) to protect the well clear volume (WCV) from intruders before they invade it. When necessary, evasive maneuvers are executed to prevent potential threats.
- **Collision Avoidance (CA):** Serving as the final line of defense, this layer is triggered when an intruder penetrates the collision avoidance threshold (CAT). CA is independent of the previous deconfliction steps and involves executing rapid and straightforward maneuvers, such as abrupt climbing/descending, to evade imminent collision.

Figure 1 illustrates the arrangement of barriers surrounding an aircraft. Beginning from the innermost boundary, the collision volume (CV) is defined by the aircraft's physical dimensions or according to aviation safety regulations. The CAT is an activation threshold for CA. The WCV is the boundary that determines the safety status of an aircraft based on the existence of other aircraft within this boundary. The outermost layer, the WCT, is designed to detect and avoid (DAA) intruders that might breach the WCV. In cases where the DAA system is not implemented, the WCT is not established. To guarantee safety, all aircraft must operate without overlapping the outermost boundary, which serves as the separation boundary. As the outermost boundary is determined by the type of deconfliction, the definition of the separation boundary changes accordingly. For example, in scenarios where only strategic deconfliction is implemented, the WCV is the separation boundary. On the other hand, in scenarios where both strategic and tactical deconfliction are implemented, the WCT is the separation boundary.

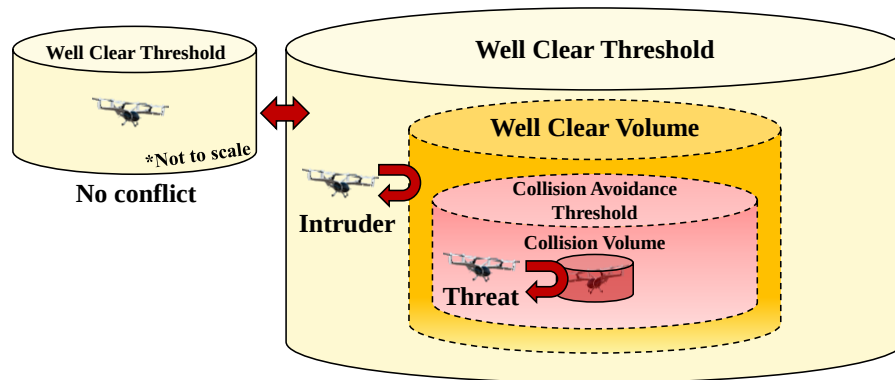


FIGURE 1 Simplified Version of Deconfliction Barriers for UAM Operations.

The size of the separation boundary can affect the traffic capacity of the constrained airspace, implying that a small separation boundary could enhance capacity. In our previous research, we introduced the methodology for determining the minimum separation boundary. The term "minimum" here refers to the smallest allowable separation boundary. The "minimum" should not be confused with the "minima" found in *separation minima* (FAA, 2019), which represents the separation boundary itself. The previous research considered scenarios where only strategic deconfliction is implemented. Building on the previous research, in this paper, we proposed the minimum separation boundary in scenarios where both strategic and tactical deconfliction are implemented. Before delving into the main part of this paper, the next section provides a brief overview of our previous research.

3.0 Safety Assured Minimum Separation Boundary

3.1 Overview of Previous Approach

Existing approaches focus on evaluating the safety risks associated with specific separation boundaries. Key approaches include the reduced vertical separation minima (RVSM) in GA (FAA, 2019), defining well clear (DWC) in UAS (Chen et al., 2019), and the building basic blocks for U-space separation (BUBBLES) risk model proposed by SESAR (Vila et al., 2023). These approaches provide a comprehensive understanding of determining separation boundaries, but heavily rely on flight data or simulations. In particular, they assume only normal operational conditions and fail to account for other operational scenarios, such as sensor failures, that could significantly increase the risk of MAC.

For the early stages of UAM, the flight data will be insufficient, making existing approaches unsuitable due to their reliance on operational data for specific separation boundaries. Moreover, given the paramount importance of safety in UAM, especially in challenging environments with high density and low altitudes, careful consideration is required when determining a tighter separation boundary. However, existing methods are inadequate for deriving minimum separation boundaries because they have focused on approximate safety assessments, thus only considering normal operations and having generalization errors across aircraft type, flight mode, equipment, etc.

To address these limitations, we proposed a systematic and conservative approach to determining the minimum separation boundary for UAM operations (Kim et al., 2023). This approach is illustrated in Figure 2, intended to be viewed from top to bottom. Our methodology involves identifying the TLS for UAM and then distributing the TLS across various subsystems of UAM. The focus is on subsystems whose uncertainty could lead to position errors, SV, and MAC, such as the navigation system. The separation boundary is a credible interval in position error. In addition, we developed an SRA that calculates the fatality (the collision risk) associated with SV. This fatality converts the allocated TLS into the SV risk requirement. The separation boundaries must be established to ensure that the probability of SV, based on the TSE distribution, remains lower than the SV risk requirement, $P_{SV,REQ}$. Then, an integrity risk analysis of each subsystem is conducted. The integrity risk analysis provides a conservative estimation of the TSE distribution while considering the operational conditions of the system. This enables our methodology to reduce its reliance on extensive flight data. Finally, the minimum separation boundary is defined as the boundary where the probability of a position error exceeding this boundary is equal to the SV risk requirement. Given that the distribution of TSE changes with the operational conditions, the minimum separation boundary is calculated for each operational condition (See Figure 2, bottom). Lastly, the largest among the boundaries is then selected as the final minimum separation boundary for UAM operations, ensuring safety across all conditions.

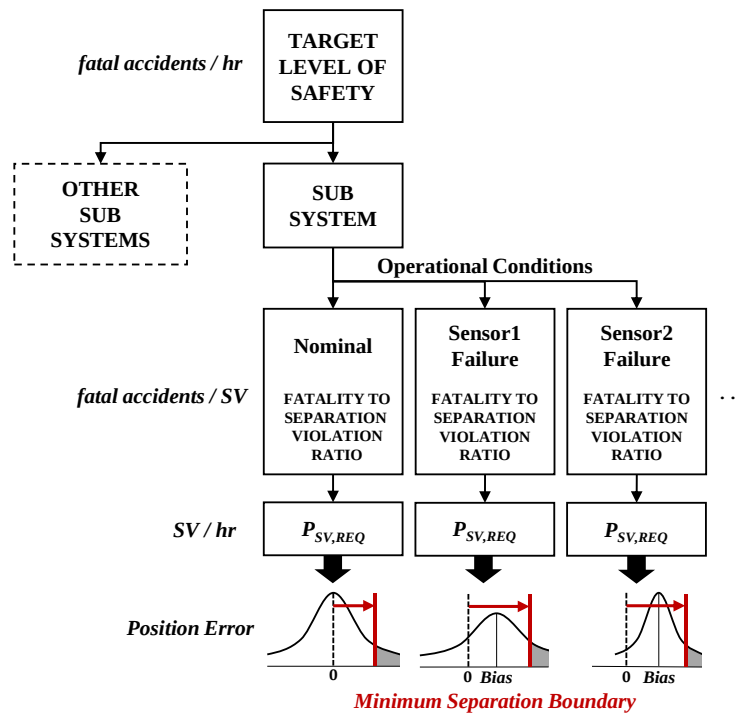


FIGURE 2 Overview of previous approach
(Adapted, with permission, from Kim et al. 2023 © IEEE proceedings).

3.2 Safety Risk Assessment for Strategic Deconfliction

The SRA assesses the fatality of SV, which is used to convert the allocated TLS to the SV risk requirement. The fatality model quantifies the probability that SV results in a series of collision and fatal accidents, represented as Equation (1):

$$P(HH) = P(HH|CO)P(CO|SV)P(SV) \quad (1)$$

where $P(HH)$ is the probability of harm to humans (HH), $P(HH|CO)$ is the probability of HH given collision (CO), $P(CO|SV)$ is the probability of CO under SV, and $P(SV)$ is the probability of SV (Kim et al., 2014). The allocated TLS is a requirement of $P(HH)$ and can be converted to the SV risk requirement according to Equation (1). $P(HH|CO)$ is assumed to be 1 given that a high likelihood of human casualties resulting from MAC. $P(CO|SV)$ represents the fatality-to-SV ratio after considering all procedures that will be performed to mitigate a potential MAC, such as the strategic and tactical deconfliction.

In our prior work (Kim et al., 2023), we calculated the fatality ratio under the assumption that no tactical deconfliction was available. As a result, the fatality became dependent on variables like traffic density, encounter geometry, and TSE distribution of engaged aircraft. To compute Equation (1) in the absence of flight data, while still maintaining a conservative approach, we introduced the following worst-case scenario using virtual traffic, as illustrated in Figure 3: First, we assume that the UAM operation was fully automated on the intended path similar to the performance-based navigation (PBN). Secondly, we assume that collisions can only be caused by the SV of the ownship ($P(CO|SV^C)=0$). This means that the virtual traffic operates normally and no intrusion into the ownship's separation boundary occurs. Thirdly, both the ownship and traffic adhere to the same corridor requirements such as TLS and performance standards. Lastly, the virtual traffic always exists on the same longitudinal plane of the ownship. To specifically assess the lateral separation boundary, the virtual traffic is placed right next to the ownship while keeping the same separation boundary as the ownship. This spatial assumption is conservative because it implies maximum traffic flow and the minimum separation boundary brings aircraft closest together.

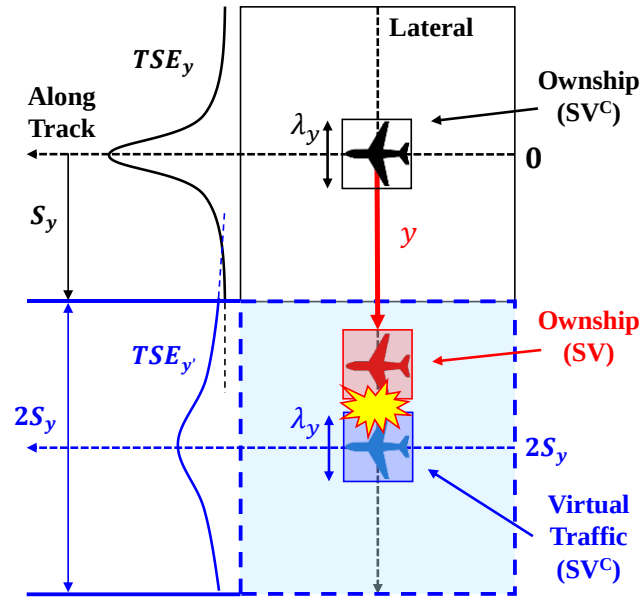


FIGURE 3 Illustration of the worst-case scenario in lateral separation. λ_y is the lateral dimension of the aircraft bounding box, S_y is the lateral dimension of the separation boundary, y is the position of the ownship and y' is the position of the traffic. TSE_y is the position error distribution of the ownship and $TSE_{y'}$ is the position error distribution of the traffic. When the ownship goes across its separation boundary ($S_y < y$) and overlaps with the normal traffic ($S_y < y' < 3S_y$), SV leads to mid-air collisions. (Adapted, with permission, from Kim et al. 2023 © IEEE proceedings).

Under this scenario, the probability of collision can be expressed by (2) for lateral separation:

$$\begin{aligned} P(CO_{NoTac}, SV) &= P(CO_{NoTac}|SV) P(SV) \\ &= 2 \int_{S_y}^{3S_y} \int_{y-2S_y-\lambda_y}^{y-2S_y+\lambda_y} g(y') \cdot f(y) dy' dy, \end{aligned} \quad (2)$$

where CO_{NoTac} is the collision without the tactical deconfliction. In Equation (2), the multiplier 2 is the number of aircraft involved in the collision, λ is the lateral dimension of the aircraft bounding box, S_y is the separation boundary with the subscript denoting the axis, $g(\cdot)$ is a TSE distribution of traffic, and $f(\cdot)$ is that of the ownship. Each pdf of TSE distribution is assumed to be Gaussian, which includes navigation system error (NSE) and flight technical error (FTE). The outer integral interval is the range in which the traffic normally exists and the inner one is the overlap range. Detailed information about the worst-case scenario and parameters can be found in (Kim et al., 2023).

We acknowledged that our model is based on several assumptions that may not align with all perspectives. Nonetheless, these assumptions make the SRA more conservative emphasizing the safety aspects. Although this approach may initially overestimate the risk, it serves as an effective and reliable strategy during the early stages of UAM, prioritizing safety above all else. As more traffic data on UAM operations becomes available, these assumptions can be adjusted in a less conservative way. The following section will introduce the methodology for determining the minimum separation boundary for the tactical deconfliction, building on the methods discussed in this section.

4.0 Minimum Separation Boundary for Tactical Deconfliction

4.1 Safety Risk Assessment for Tactical Deconfliction

The tactical deconfliction employs the DAA system to reduce the risk of MAC when the strategic deconfliction fails and SV occurs. The DAA system detects intruders who may have or already have violated the separation boundary and initiates an evasive maneuver if necessary. Therefore, detection and avoidance performance are two key parameters of the DAA system. Detection performance depends on the surveillance system and the detection algorithm (Kochenderfer et al., 2012), while avoidance performance is influenced by the encounter geometry, relative velocity at the start of the avoidance, aircraft maneuverability, and the avoidance algorithm (Datta & Oliver, 1992; Erzberger et al., 1997). Although the tactical deconfliction is effective at reducing collision risks, collisions can still occur if the detection fails or if, despite correct detection, the avoidance fails. This scenario is illustrated in Figure 4.

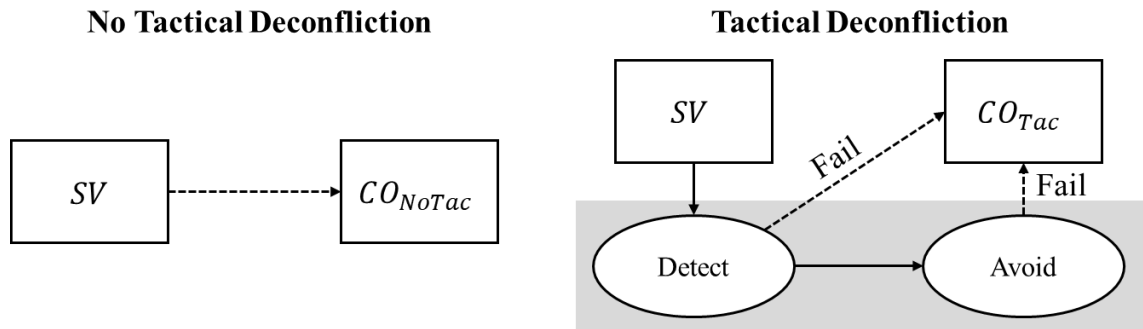


FIGURE 4 Comparison of the collision scenarios with and without the tactical deconfliction. SV is a separation violation, CO_{NoTac} is a collision without the tactical deconfliction, and CO_{Tac} is a collision with the tactical deconfliction. The tactical deconfliction reduces the risk of collision by utilizing a detect-and-avoid system.

The probability of HH resulting from SV and the subsequent failure of the DAA system is expressed as the sum of the collision probabilities of two scenarios: the collision under correct detection (Det) and the collision under detection failure ($DetF$). Therefore, Equation (1) can be reformulated as follows:

$$P(HH) = P(HH|CO_{Tac}) \cdot \left(\frac{P(CO_{Tac}|Det, SV) P(Det|SV)}{+P(CO_{Tac}|DetF, SV) P(DetF|SV)} \right) \cdot P(SV), \quad (3)$$

where CO_{Tac} is the collision with the tactical deconfliction. In the equation, $P(HH|CO_{Tac})$ is conservatively set to one, and the sum of $P(Det|SV)$ and $P(DetF|SV)$ equals one. $P(CO_{Tac}|Det, SV)$ is the probability of the collision given that SV is correctly detected. This reflects the scenario where an avoidance maneuver is executed following correct detection but ends up in a collision. In contrast, $P(CO_{Tac}|DetF, SV)$ is the probability of collision given that SV is not detected and consequently, no avoidance maneuver is executed. Hence, this probability is the same as the collision probability in scenarios without the tactical deconfliction, where the avoidance maneuver is not implemented either:

$$P(CO_{Tac}|DetF, SV) = P(CO_{NoTac}|SV). \quad (4)$$

Using Equation (3) and substituting $P(HH|CO_{Tac})$ with one, and $P(DetF|SV)$ with $1-P(Det|SV)$, the probability of HH is reformulated as

$$\begin{aligned}
P(HH) &= 1 \cdot \left(\frac{P(CO_{Tac}|Det,SV) P(Det|SV)}{+P(CO_{Tac}|DetF,SV) P(DetF|SV)} \right) \cdot P(SV) \\
&= \left(\frac{\varepsilon \cdot P(CO_{NoTac}|SV) P(Det|SV)}{+P(CO_{NoTac}|SV) (1 - P(Det|SV))} \right) \cdot P(SV) \\
&= \left(\frac{\varepsilon \cdot P(Det|SV)}{+(1 - P(Det|SV))} \right) \cdot P(CO_{NoTac}|SV) P(SV) \\
&\text{where } \varepsilon = \frac{P(CO_{Tac}|Det,SV)}{P(CO_{NoTac}|SV)}.
\end{aligned} \tag{5}$$

The probability of HH is now modeled as a function of three variables. The first is $P(CO_{NoTac},SV)$, the collision probability in the absence of the tactical deconfliction, as previously described in Section 2. The second is $P(Det|SV)$, the probability of detection given SV, which is related to the DAA system's detection performance. The third parameter is the ratio of $P(CO_{Tac}|Det,SV)$ to $P(CO_{NoTac}|SV)$, which we newly define and call the "avoidance risk ratio (ARR)." ARR serves as an indicator of the DAA system's avoidance performance. The ratio represents the reduction in the collision risk achieved by the avoidance maneuver compared to scenarios without any maneuver. A lower ARR indicates a more effective avoidance maneuver, whereas an ARR exceeding one indicates that the risk of collision is potentially higher than not taking the maneuver and leaving it to sheer luck.

This approach is akin to a concept in NASA's report on defining WCV in UAS (Chen et al., 2019). This report introduces the "risk ratio (RR)" as the probability ratio of near mid-air collisions (NMAC) with and without the DAA system. While RR evaluates the overall performance of the DAA system, ARR specifically assesses the performance of its avoidance part. This difference allows for a more detailed analysis of the DAA system. Using ARR along with the detection probability allows us to quantify the risk associated with detection and the avoidance maneuver, separately. Our approach analyzes each configuration of the DAA system, such as the surveillance system and avoidance algorithm, thus facilitating the detailed standardization of the DAA system. This is particularly advantageous in the early stages of UAM.

In comparison to Equation (1), which calculates the collision probability without the tactical deconfliction, Equation (5) includes factors that potentially reduce the collision risk through the term offered by the DAA system. If SV is detected, the collision risk is reduced by a factor of ARR. Conversely, if SV is not detected, the fatality risk remains as it would be without the tactical deconfliction. When the probability of detection is zero, Equation (5) is simplified to Equation (1). In conclusion, to lower the safety risk using the tactical deconfliction, the DAA performance must first ensure that ARR is less than one. Once this condition is met, the safety risk can be further reduced as the probability of detection approaches 1.

4.2 Determining the Minimum Separation Boundary for Tactical Deconfliction

In this paper, our focus is on the nominal condition (see Figure 1, bottom left) where the TSE distribution is unbiased and follows a zero-mean Gaussian distribution. As discussed in Section 3, the minimum separation boundary is defined as the point where the safety risk aligns with the allocated TLS. The one-dimensional (1-D) search method is used to determine the minimum separation boundary at which the probability of HH matches to the allocated TLS. This approach is necessary because there is no analytic solution for the separation boundary in Equation (2). Additional information on the failure conditions and the search method can be found in (Kim et al., 2023).

For the tactical deconfliction, the minimum separation boundary is derived from Equations (2) and (5). On the other hand, the boundary for scenarios without the tactical deconfliction is determined solely by Equation (5). The term offered by the DAA system in Equation (5), which is less than one, can relax the safety requirement because the allocated TLS is divided by this term. This relaxation of the allocated TLS could provide a margin that allows the boundary to be reduced while still meeting the same requirement. As a result, a further reduction in the separation boundary could be achieved compared to scenarios that rely solely on the strategic deconfliction. The following section applies the proposed methodology to ACAS-X, an advanced DAA system, as a case study to explore the performance parameters of the DAA system.

5.0 Case Study

5.1 ACAS-X Detection Algorithm

In the context of UAM, transitioning to advanced DAA systems such as ACAS-X is essential for ensuring safety within the increasingly congested airspace (FAA, 2020; EUROCONTROL, 2022). The traffic collision avoidance system (TCAS), a conventional DAA system mandated for large aircraft in most countries, scans the airspace and issues alerts to prevent collisions. However, TCAS face challenges in meeting the unique demands of UAM because it relies on a beacon-based surveillance system that includes a Mode-S transponder without GPS integration. This system is expensive, power-intensive for small aircraft, and offers limited performance due to the absence of GPS (Kochenderfer et al., 2012). ACAS-X improves both cost efficiency and reliability by integrating the automatic dependent surveillance-broadcast (ADS-B) (EUROCONTROL, 2022). ADS-B is an enhanced surveillance system that leverages GPS within the Mode-S transponder framework. It periodically broadcasts an aircraft's position, altitude, and velocity derived from GPS. This enables aircraft to utilize reliable state information transmitted by other equipped aircraft for preventing collision.

Figure 5 illustrates a simplified structure of the ACAS-X that is analyzed in this case study. The system utilizes an ADS-B receiver to monitor the state of approaching aircraft and a dedicated GPS that operates independently from the main navigation system to estimate the state of ownship. This configuration provides cost-effective and reliable estimates for both states. Subsequently, the ACAS-X applies the TCAS detection logic to determine whether the approaching aircraft is an intruder.

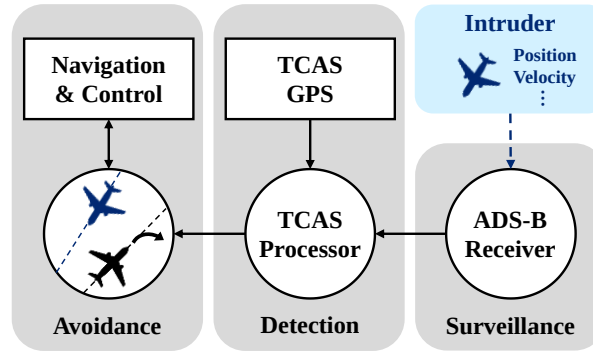


FIGURE 5 Simplified structure of ACAS-X for UAM operations. This configuration includes an ADS-B receiver for enhanced situational awareness, TCAS detection logic, and a dedicated GPS that operates independently of the main navigation system.

The TCAS detection logic flags an approaching aircraft as an intruder if its closest point of approach (CPA) violates the separation boundary and the remaining time to CPA is less than a predetermined threshold (Jamoon et al., 2016). This logic can be expressed as the following equation:

$$Det = [\hat{t} \leq \tau_{THR} \cap \hat{r}_{CPA} \leq S_y \cap \hat{z}_{CPA} \leq S_z] \quad (6)$$

Here, $\hat{t}$ is the estimated time remaining to the CPA, $\hat{r}_{CPA}$ and $\hat{z}_{CPA}$ are the estimated horizontal and vertical distances from the ownship to the CPA, respectively. τ_{THR} is the threshold for the time remaining to CPA, and S_y and S_z represent the horizontal and vertical sizes of the separation boundary. These estimates, called hazard states, are relative values between the two aircraft and are considered independent of each other (Jamoon et al., 2016). In ACAS X, they are determined by ADS-B, which is subject to uncertainty, potentially leading to detection failures as illustrated in Figure 6 (left).

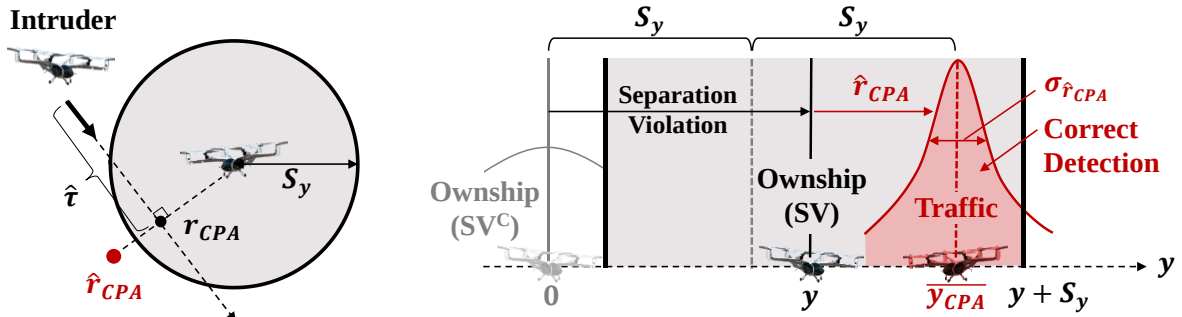


FIGURE 6 Detection failure due to uncertainty in the surveillance system (left) and the worst-case scenario for a one-dimensional lateral case (right). Note that y originates from zero, while r is a relative value that uses the ownship as its starting point.

Figure 6 (left) illustrates the application of the TCAS detection logic under the worst-case scenario discussed in Section 3. For simplicity, the discussion is limited to a one-dimensional lateral scenario, with both $\hat{r}$ and $\hat{z}_{CPA}$ meeting the detection conditions in Equation 6. When the SV occurs at the ownship, it crosses its separation boundary and moves toward the traffic. The DAA system perceives this as the traffic approaching the ownship. The surveillance system estimates the hazard states, but carries uncertainty. The estimated relative distance, $\hat{r}_{CPA}$, is modeled as a Gaussian distribution with a mean $\overline{y_{CPA}} - y$ and variance $\sigma_{\hat{r}_{CPA}}$. Here, $\overline{y_{CPA}}$ is the average distance to the traffic in the normal condition (SV^C), which is $2S_y$, and $\sigma_{\hat{r}_{CPA}}$ is the surveillance uncertainty. Specifically, $\sigma_{\hat{r}_{CPA}}$ serves as an important indicator of the detection performance and is selected as a key variable in the following sensitivity analysis. Consequently, the probability of correct detection can be expressed as follows:

$$\begin{aligned}
P(Det, SV) &= P(\hat{r}_{CPA} \leq S_y | \overline{y_{CPA}} < y + S_y) P(\overline{y_{CPA}} < y + S_y). \\
&= P(\hat{r}_{CPA} \leq S_y | S_y < y) P(S_y < y) \quad (\because \overline{y_{CPA}} = 2S_y) \\
&= P(\hat{r}_{CPA} \leq S_y | SV) P(SV) \\
&= \int_{S_y}^{3S_y} \int_{y-S_y}^{y+S_y} h(u) \cdot f(y) du dy,
\end{aligned} \tag{7}$$

where u is the estimated distance to the traffic and $h(\cdot)$ is the Gaussian distribution with a mean $\overline{y_{CPA}}$ and variance $\sigma_{\hat{r}_{CPA}}$. The outer integral interval is the range where a collision with normal traffic could occur and the inner one is the detection region.

To summarize, the surveillance uncertainty is defined as an indicator of the detection performance. This uncertainty could be influenced by factors such as ADS-B type, weather conditions, and even the distance between aircraft, although the latter is not considered in this study. Finally, the term offered by the DAA system in Equation (5) becomes a function of two variables: the surveillance uncertainty and ARR. In addition to the surveillance uncertainty, ARR is also the key variable in the sensitivity analysis. However, in this study, ARR is analyzed directly without applying any specific model due to the absence of established models. The development and application of ARR modeling is an ongoing area of our research.

5.2 Sensitivity Analysis

Sensitivity analysis assesses the impact of surveillance uncertainty and ARR on the minimum separation boundary. Table I summarizes the parameters used in this analysis. In this analysis, we assumed the nominal condition of the ownship's subsystem including the DAA system, and thus the ownship's TSE and the hazard state follows a zero-mean Gaussian distribution. The variance of the TSE is calculated as the root sum squared of NSE and FTE. The ownship's NSE is estimated by the navigation system, utilizing GPS L1 and consumer-grade inertial measurement unit (IMU) (Groves, 2013).

TABLE 1

Parameters for the sensitivity analysis

Parameter		Value
Target Level of Safety ($TLS_{Nav,Nom}$)		10^{-7} Fatal accidents/hr
Ownship Accuracy (σ_{TSEy})	Navigation System Error	27.6 m
	Flight Technical Error	15 m
95% Traffic Accuracy ($2 \cdot \sigma_{TSEy'}$)		185.2 m
Max. Lateral dimension of aircraft (λ_y)		14 m

In this study, the concept of corridor requirements is introduced, setting some parameters to their maximum allowable values. Blue boxes in Table 1 indicate the corridor requirements. The TLS of the navigation system in a nominal condition (see Figure 1, left column) is 10^{-7} fatal accidents per flight hour, an example of safety requirement associated with navigation systems in GA. According to PBN, the 95% accuracy requirement is defined as the capability to maintain a certain distance from the intended path (ICAO, 2008). This analysis adopts the RNP 0.1 specification, representing the most stringent accuracy requirement under PBN guidelines. The specification has also been used in analyses like those by Zahn and Ringelberg, 2023. The FTE requirement is 15 m, which is the FTE requirement for level flight below 1,500 m in PBN. The lateral dimensions of both the traffic and ownship aircraft is set at 14 m (46.10 ft), which is the wingspan of the Joby S4 airframe (Zahn & Ringelberg, 2023). Lastly, the TSE distribution of the traffic is a zero-mean Gaussian with variance equal to the accuracy requirement.

The 1-D search process begins with a large separation boundary and gradually decreases it down, assessing the safety risk associated with the separation boundary as per Equation (5). As the separation boundary decreases, the safety risk increases, and the boundary whose safety risk aligns with the allocated TLS becomes the minimum separation boundary.

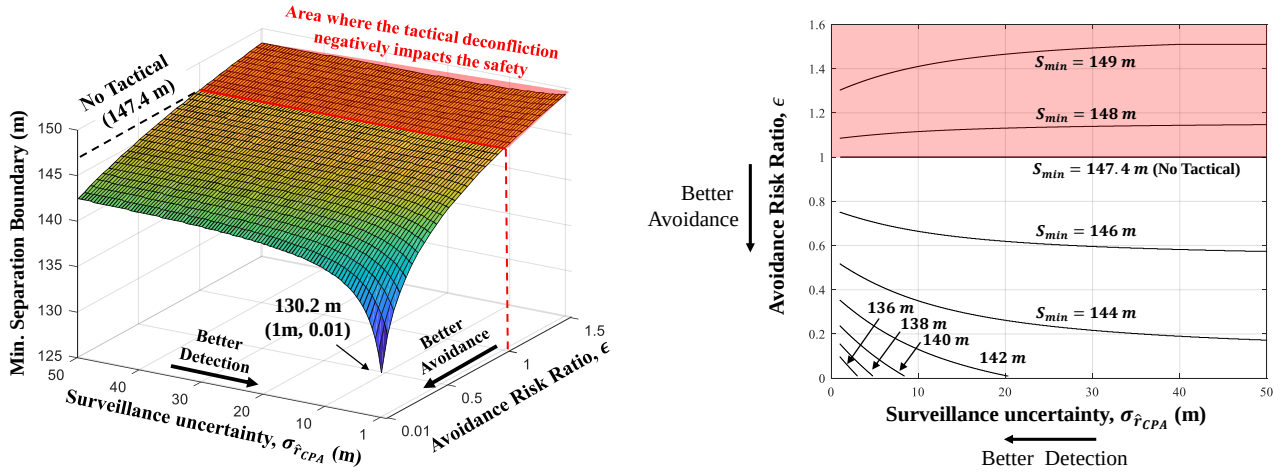


FIGURE 7 Result of the sensitivity analysis: Impact of surveillance uncertainty and avoidance risk ratio on the minimum separation boundary.

The results from the sensitivity analysis are shown in Figure 7, which presents the impact of the surveillance uncertainty and ARR on the minimum lateral separation boundary. Here, the x-axis represents the surveillance uncertainty, ranging from 1 to 50 m, and the y-axis represents ARR, ranging from 0.01 to 1.5. Lower values of both variables indicate improved performance. In the figure, we can observe that the minimum separation boundary decreases as both variables decrease. Specifically, a surveillance uncertainty of 1 meter and an ARR of 0.01 represent the practical minimums in this study. These values can be achieved by augmenting the ACAS X with surveillance systems employing real-time kinematics (RTK) (Groves, 2013) and avoidance algorithms such as the detect and avoid alerting logic for unmanned systems (DAIDALUS) (Cook et al., 2015). This configuration results in a separation boundary reduction to 130.2 meters, which is an 11.7% enhancement compared to 147.4 meters for the baseline scenario without the tactical deconfliction. Moreover, improvements in surveillance uncertainty and ARR beyond the practical minimum can further reduce the minimum separation boundary, potentially reaching below 130.2 meters.

Figure 7 (right) analyzes the performance improvements required for every one- or two-meter decrease in the separation boundary. It demonstrates that reducing both variables simultaneously accelerates the reduction. Given the high sensitivity of the minimum separation boundary to ARR near the baseline, the significance of surveillance uncertainty reduction will remain obscured until there is a substantial enhancement in ARR. Therefore, in the early stages of UAM, where the DAA system is under development, it is effective to prioritize improving avoidance algorithm. Additional reductions in the minimum separation boundary can be achieved by enhancing the navigation system through multi-constellation, multi-frequency, and IMU integration or by imposing more stringent corridor requirements, such as FTE and accuracy requirements (Kim et al., 2023).

However, as shown in the red region of Figure 7 (right), an ARR greater than one results in a separation boundary that exceeds the baseline scenario, regardless of detection performance. This indicates that the tactical deconfliction could adversely affect safety. Therefore, avoidance algorithms should maintain an ARR below one to enhance safety and reduce the separation boundary. A notable observation in the red area of Figure 7 (left) is that lower surveillance performance paradoxically appears to enhance safety, resulting in a smaller separation boundary. This phenomenon could be intuitively explained by a strategy of inaction, relying on sheer luck rather than engaging in potentially hazardous maneuvers.

This study acknowledges the need for further research, particularly in accounting for potential subsystem failures, including the navigation and DAA systems, to ensure high safety levels under all operational conditions (Kim et al., 2023). Moreover, the effect of ARR in this study does not fully consider the relationship with the separation boundary, given that a certain separation is necessary to successfully execute evasive maneuvers. Future work will address these topics.

6.0 Conclusion

In urban air mobility (UAM) operations, constrained airspace, stringent safety requirements, and a lack of flight data to validate safety have presented a new challenge in determining the size of the separation boundary. To tackle these challenges, our prior research introduced the concept of a minimum separation boundary that maximizes airspace efficiency while still satisfying the safety requirement. It proposed a methodology for determining the minimum separation boundary with reduced reliance on flight data, and developed a novel safety risk assessment (SRA) to evaluate the collision probability in scenarios where the strategic deconfliction is implemented. Building on this foundation, the current study expands the scope of SRA to include the tactical deconfliction, which utilizes a detect-and avoid (DAA) system to additionally reduce collision risks when the strategic deconfliction fails. The integration of the DAA system has the potential to enhance safety, allowing for a further reduction in the separation boundary. In this context, our study proposes an improved SRA that quantifies the safety margin provided by the tactical deconfliction, and defines detection probability and avoidance risk ratio (ARR) as key performance parameters for the DAA system.

The proposed methodology was applied to airborne collision avoidance system X (ACAS X) as a case study. In this case study, the detection probability was modeled as a function of surveillance uncertainty. Consequently, the detection and avoidance performance of ACAS X are parameterized by the surveillance uncertainty and ARR, respectively. A sensitivity analysis was conducted to analyze the impact of the surveillance uncertainty and ARR on the minimum separation boundary. The sensitivity analysis revealed that implementing the tactical deconfliction could reduce the minimum separation boundary by 11.7% compared to scenarios without it. Furthermore, our research has demonstrated conditions under which the DAA system could compromise the system safety, and has identified a performance criterion: the ARR of avoidance algorithms should be less than one. The methodology developed through this study would provide a practical framework for UAM stakeholders to explore the effective use of airspace and the design of the DAA system, particularly during the early stages of UAM.

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